

The **Library Management System API** is a RESTful API built with Django REST Framework that allows managing books, authors, library members, and borrowing/returning operations.

books (add, update, delete, view list, view details)

authors (add, update, delete, view)

members (add, update, delete, view)

borrow records (which member borrowed which book, and return status)

availability status of books is updated on borrow/return

filtering, searching, pagination for easy access

Database Schema (Models Definition)

Author

Field	Type	Description
id	Primary Key	Unique identifier for the author
name	CharField	Author's name
biography	TextField	Short bio of the author

Field	Type	Description
create_at	DateTime	Record creation time
updated_at	DateTime	Record update time

Book:

Field	Type	Description
id	Primary Key	Unique identifier for the Book
<u>title</u>	<u>CharField</u>	Book title
Author	ForeignKey	Book Author
ISBN	CharField(unique)	ISBN number
Category	CharField	Book category
availability_status	ChoiceField	Book availability
Book_pages	DecimalField	Number of pages
price	DecimalField	Price of the book

Member:-

<u>Field</u>	<u>Type</u>	<u>Description</u>
<u>id</u>	<u>AutoField</u> (PK)	Unique identifier for the member
name	<u>CharField</u>	Member's name
<u>email</u>	<u>EmailField</u>	Unique email
<u>membership_date</u>	<u>DateField</u>	Membership start date
<u>create_at</u>	<u>DateTime</u>	Record creation time
<u>updated_at</u>	<u>DateTime</u>	Record update time

BorrowRecord:-

<u>Field</u>	<u>Type</u>	<u>Description</u>
<u>id</u>	AutoField (PK)	Unique identifier for the record
<u>book</u>	ForeignKey	Borrowed book
<u>member</u>	ForeignKey	<u>Borrower</u>

<u>Field</u>	<u>Type</u>	<u>Description</u>
<u>borrow_date</u>	<u>DateTime</u>	<u>Borrowing date</u>
<u>return_date</u>	<u>DateTime</u>	Returning date
<u>status</u>	<u>ChoiceField</u>	Current borrow status

API Endpoints Definition:-

Books:-

GET /books/ Retrieve all books

POST /books/ Add a new book

GET /books/{id}/ Retrieve book details

PUT /books/{id}/ Update book details

DELETE /books/{id}/ Remove a book

Authors:-

- GET /authors/Retrieve all authors
- POST /authors/ Add a new author
- PUT /authors/{id}/ Update author details
- DELETE /authors/{id}/ Remove an author

Members

- **GET /members/ Retrieve all members**
- **POST /members/ Add a new member**
- **PUT /members/{id}/ Update member details**
- **DELETE /members/{id}/ Remove a member**

Borrow Records

- **GET /borrow-records/ Retrieve all borrow records**
- **POST /borrow/ Borrow a book (create borrow record)**
- **POST /return/ Return a book (update borrow record status)**